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ALZHEIMER'S PREVENTION INITIATIVE

An Examination of Rate of Decline as an Alternative to Change from Baseline

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Disclosures

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NOT — LU ET AL
NOTES — HU, MACKEY, THOMAS

Outline

- Change (Δ) from baseline summary measure[1] and use of Mixed Model Repeated Measures (MMRM)
- Rate of decline summary measure and use of Random Coefficient Regression Model (RCRM)
- Other notable work
- Advantages/disadvantages of each
- Conclusion

[1]Model, AD Scientific Working Group CTAD 2017

Change from baseline (Δ) at time T via MMRM

H_0: Δ from baseline at time T (ΔT) in control group = ΔT in experimental group

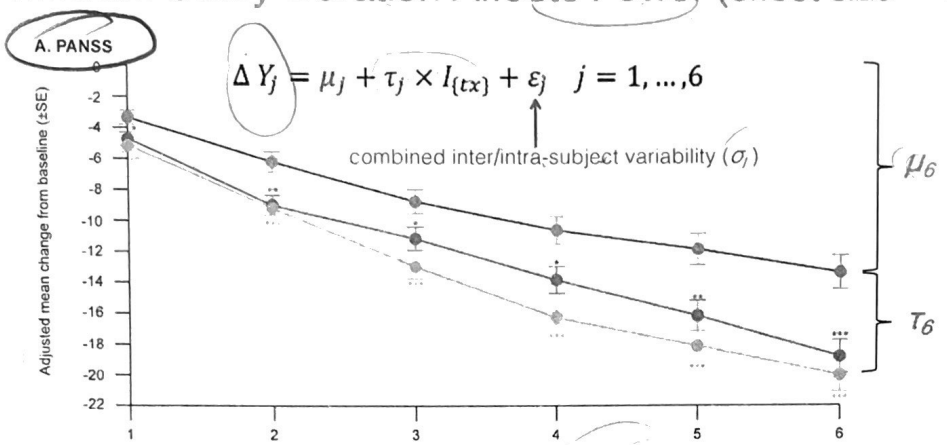
- Efficacy at time T only
- Time modeled on visit scale (i.e. effect at visit x)
  - time's effect on endpoint is left unspecified
- Inter- & intra-subject variation not modeled separately
- Regulatory approvals (e.g. FDA, EMA, etc.) based on MMRM

6 week efficacy of brexpiprazole in patients with acute schizophrenia[1], approved 2015

Study	Treatment Group	N	Primary Efficacy Measure: PANSS		
			Mean Baseline Score (SD)	LS Mean Change from Baseline (SE)	Placebo-subtracted Difference[a] (95% CI)
3	REXULTI (2 mg/day)*	180	95.9 (13.8)	-20.7 (1.5)	-8.7 (-13.1, -4.4)
	REXULTI (4 mg/day)*	178	94.7 (12.1)	-19.7 (1.5)	-7.6 (-12.0, -3.1)
	Placebo	178	95.7 (11.5)	-12.0 (1.6)	--

[1]Correll et al. Schizophr Res 2016,174 82–92

MMRM: Study Duration Affects Power (effect size = τ_6/σ_6)



- Provides 'robust' estimate of efficacy at wk 6
- Measurements outside of wk 0 & wk 6 provide little information about H_0 or H_A
- Power affected by study length, # of assessments has little impact
- Patients w/1 assessment not included; unscheduled visits?

[1]Correll et al. Schizophr Res 2016,174 82–92

Rate of decline via RCRM*

H_0: rate of decline in control group = rate of decline in the experimental group
or H_0: f(time) = g(time)

- Efficacy summarized over entire trial, highly efficient use of all data[1]
- Time modeled directly
  - often linearly (i.e. constant rate of decline)
- Inter- & intra-subject variation modeled separately
- Regulatory approval based on RCRM

Effect of nintedanib on rate of decline in patients with idiopathic pulmonary fibrosis[2], approved 2014

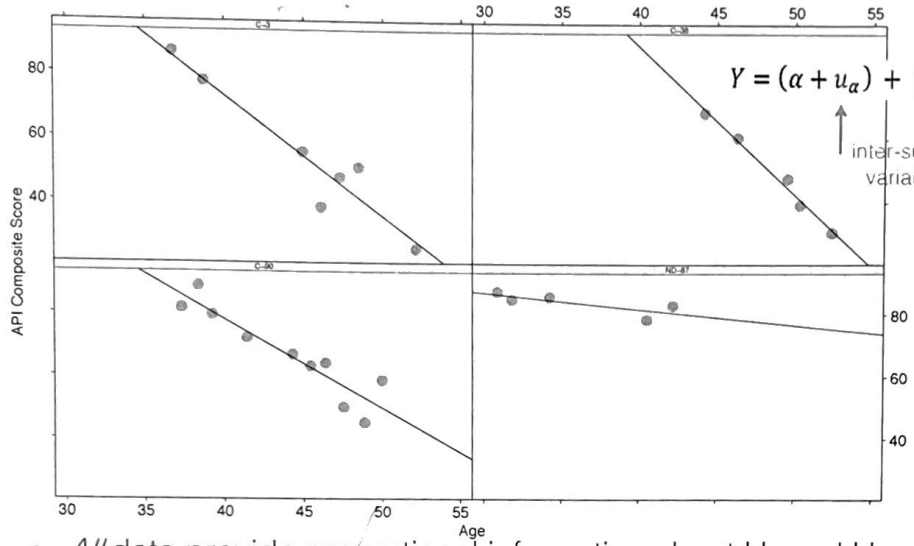
	Study 1		Study 2		Study 3	
	OFEV 150 mg twice daily	Placebo	OFEV 150 mg twice daily	Placebo	OFEV 150 mg twice daily	Placebo
Number of analyzed patients	84	83	309	204	329	219
Rate[a] of decline over 52 weeks	-60	-191	-115	-240	-114	-207
Comparison vs placebo Difference[b]	131		125		94	
95% CI	(27, 235)		(78, 173)		(45, 143)	

[1]Lu Biometrics 2010, 66, 891–896 [2]NEJM 2014 370 82–92
*Also known as random effects model, mixed effects model, constrained longitudinal analysis model, or continuous time MMRM

WHAT IS INTERACTION?

RESIDUAL VAR

RCRM: Duration, #, and Placement of Assessments Affect Power



$$Y = (\alpha + u_\alpha) + (\beta + u_\beta) \cdot time + \varepsilon$$

inter-subject variability

intra-subject variability

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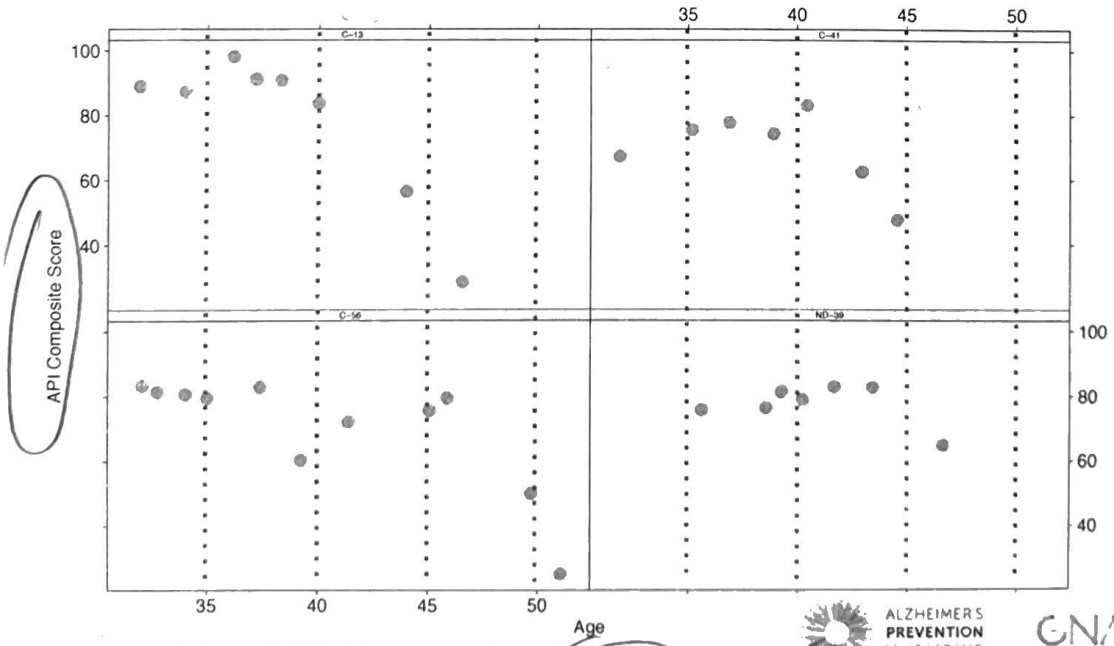
- *All* data provide proportional information about H_0 and H_A
- Assessment times can be used as a tool to optimize efficiency

Ayutyanont et al. J Clin Psychiatry 2014;75:652-60. Rios-Romenets et al. Alzheimer's & Demntia 2016;13:602-05



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Ayutyanont et. al. J Clin Psychiatry 2014;75:652-60. Rios-Romenets et al. Alzheimer's & Demntia 2016;13:602-05

Partial list of notable work

- *Mixed Model of Repeated Measures Versus Slope Models in Alzheimer's Disease Clinical Trials* (2012)[1]
  - AIC model selection favored RCRM over MMRM in 5 AD clinical trials
- *A Simulation Study Comparing Slope Analysis with Endpoint Analysis for Analyzing Cognition Data in Clinical Trials of Alzheimer's Disease* (2014)[2]
  - ≥13% power improvement from RCRM over MMRM
  - Type I error was controlled, even under severe non-linearity → robust for testing H_0
- *Longitudinal decline in mild-to-moderate Alzheimer's disease: Analyses of placebo data from clinical trials* (2016)[3]
  - Linear to near linear decline observed in 20 mild-to-moderate AD trials

[1]Donohue & Aisen. J Nutr Health Aging 2012,16:360–4, [2]Chen et al CTAD 2014, [3]Thomas et al Alzheimers Dement 2016,12:598–603

MMRM[1,2] Power

$$\Phi\left(\frac{\beta_{tx}\sqrt{\frac{mn}{m+n}}}{\sqrt{\frac{\sigma^4+2\sigma^2\sigma_0^2}{(\sigma_0^2+\sigma^2)T^2}+\frac{2\sigma_{01}\sigma^2}{(\sigma_0^2+\sigma^2)T}+\frac{\sigma_0^2\sigma_1^2+\sigma^2\sigma_1^2-\sigma_{01}^2}{(\sigma_0^2+\sigma^2)}}}-z_{0.975}\right)$$

- Longer studies increase power
- Homogeneous populations can be studied with smaller sample sizes if long enough
- Heterogeneous populations get diminishing benefit form longer studies
- Straightforward to calculate power for various study durations

[1]Lu et al. Statistics in Medicine 2000; 28:679-693
[2]Hu, Mackey, Thomas. In preparation 2018

Relative Reduction (RR) by RCRM = RR by MMRM

Efficacy in AD phase 3 trials is commonly summarized with RR at time T

- $RR_{MMRM} = \frac{\text{Placebo decline} - \text{treatment decline}}{\text{Placebo decline}}$
- $RR_{RCRM} = \frac{\text{Placebo rate of decline} - \text{treatment rate of decline}}{\text{Placebo rate of decline}}$
- Under linearity, $RR_{MMRM} = RR_{RCRM}$

RR_{RCRM} analogous to Hazard Ratio used in Oncology & Cardio

In randomized oncology/cardiology studies, H_0: f(time) = g(time)

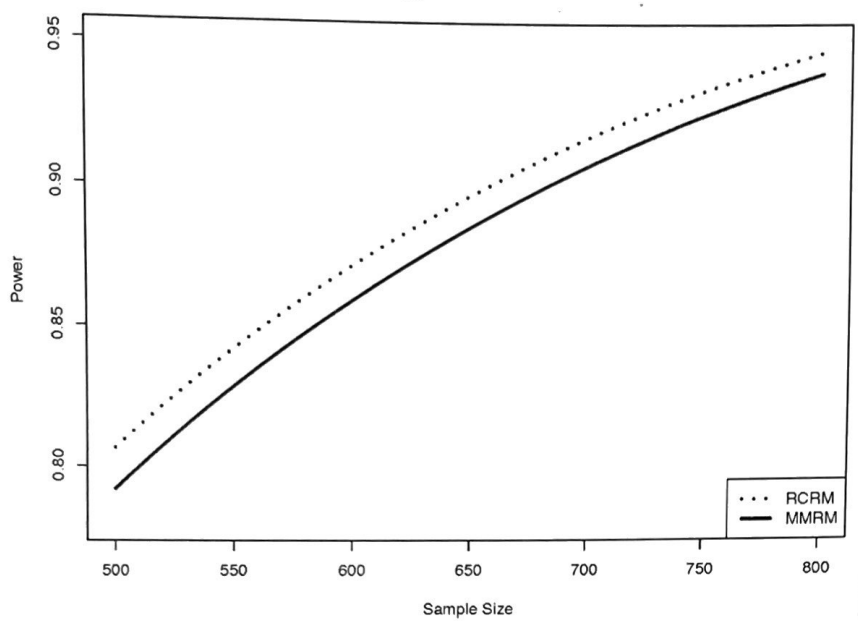
- Efficacy is summarized using the Cox Proportional Hazards model
- Estimates
  - ratio of hazard* rates when the ratio of rates is constant over time
  - an average ratio of hazards when ratio is not constant over time

Hazard ratio = 0.6 → "Tx A reduces risk of death by 40% (= 1 - 0.6)"

Rate of decline = 0.6 in tx & 1 in control
→ "Tx B reduces the rate of decline by 40%"

*a measure of risk the event (death or progression)

RCRM: 1% power gain over MMRM[1] in this mild AD trial[2,3]

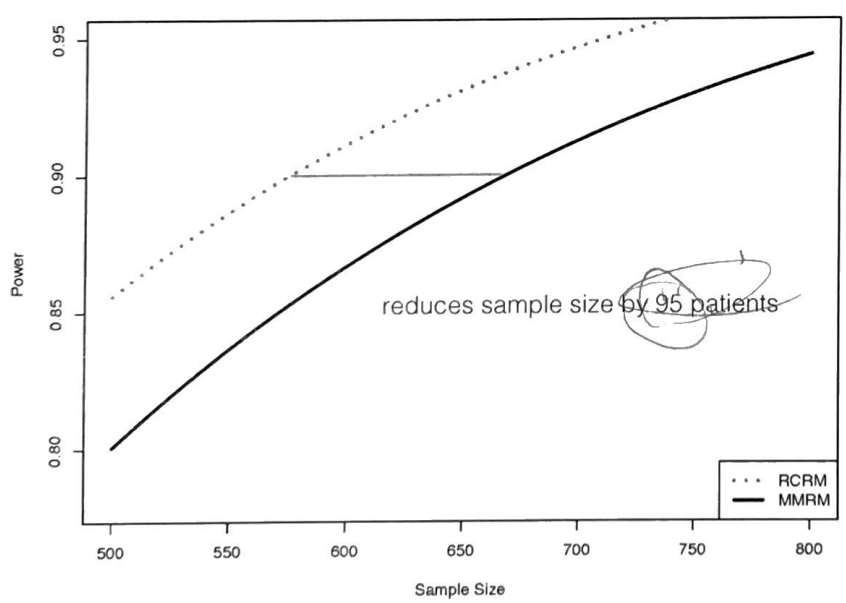


What would happen if we moved the 2nd assessment 3 months earlier and the 3rd assessment 3 months later?

- Type I error = 0.05
- 30% RR
- 1:1 randomization
- 10% dropout per yr
- 1.5 yr follow-up
- assessments every 6 months
- longitudinal MMRM[2] (slightly more powerful than Δ from baseline MMRM)
- random intercept SD = 1.4
- random slope CV= 0.99
- intercept-slope covariance =1.3
- intra-subject SD = 1.1

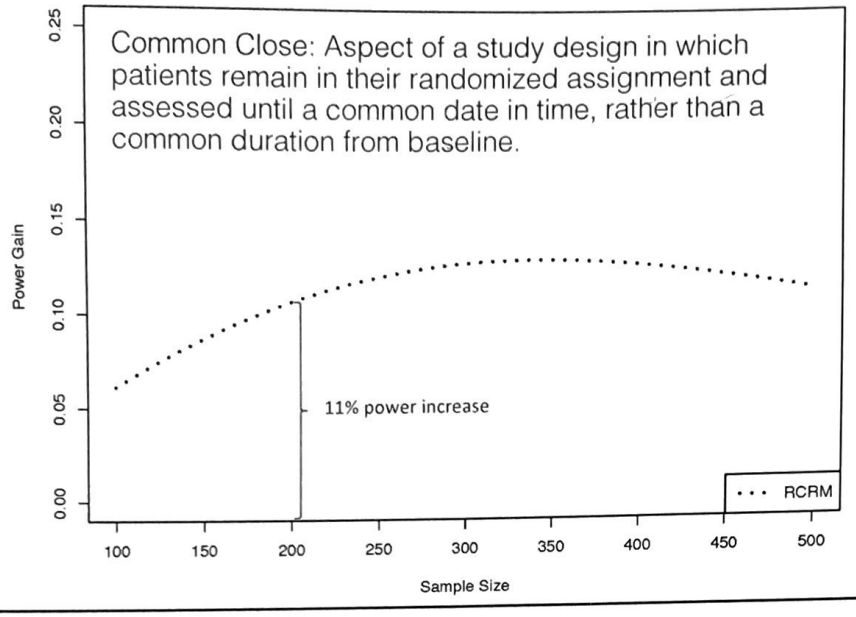
[1]Lu et al. Statistics in Medicine 2009, 28:679–699
[2]Lu Biometrics 2010, 66, 891–896
[3]Hu, Mackey, Thomas In preparation 2018

RCRM: Additional 4% improvement by moving 2 visits



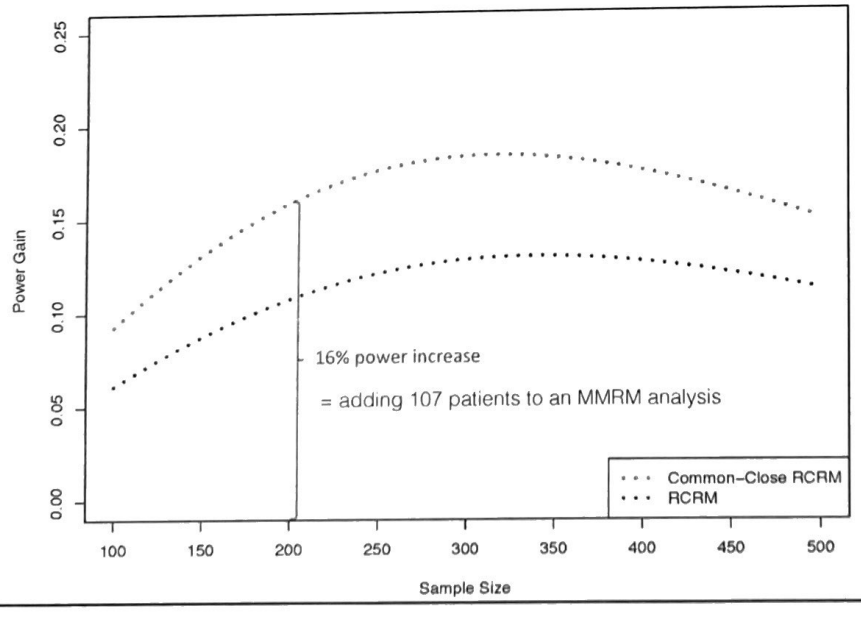
- visits @ baseline, 3, 15, & 18 months

RCRM: substantial power ↑ over MMRM in an ADAD trial



- Type I error = 0.05
- 30% RR
- 1:1 randomization
- 5% dropout per yr
- 5 yr study
- visits every 6 months
- longitudinal MMRM
- random intercept SD = 6.8
- random slope CV = 0.86
- intercept-slope covariance = 2.7
- intra-subject SD = 1.2

RCRM: Common close provides additional power boost



- 36 month enrolment period

Conclusions

Under linearity, $RR_{MMRM} = RR_{RCRM}$, analogous to oncology/cardio trials

MMRM

- Robust in estimating $\Delta\tau$
- Doesn't efficiently leverage all data
- Cannot benefit from common close

RCRM

- Likely robust for H_0: $f(t) = g(t)^*$, however average RR for non-constant decline rates
- Under linearity & > 1 post-baseline assessment, always more powerful than MMRM
- Use of common close or optimization of assessment timing adds additional power
- May be particularly powerful in small studies in homogeneous populations

*requires further research to confirm

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BACKUP

MMRM[1] & RCRM[2] POWER

$$MMRM: \Phi\left(\frac{\beta_{tx}\sqrt{\frac{mn}{m+n}}}{\sqrt{\frac{\sigma^4+\sigma_0^2\sigma_1^2T^2+\sigma^2\sigma_1^2T^2-\sigma_{01}^2T^2+2\sigma^2\sigma_0^2+2\sigma_{01}\sigma^2T}{(\sigma_0^2+\sigma^2)T^2}}}-z_{0.975}\right)$$

$$RCRM: \Phi\left(\frac{\beta_{tx}\sqrt{\frac{mn}{m+n}}}{\sqrt{\frac{\sigma^4+(J+1)\sigma^2\left(\sigma_1^2\overline{t^2}+\sigma_0^2+2\sigma_{01}\bar{t}\right)+(J+1)^2(\sigma_0^2\sigma_1^2-\sigma_{01}^2)\left(\overline{t^2}-\bar{t}\cdot\bar{t}\right)}{(J+1)\left(\sigma^2\overline{t^2}+(J+1)\sigma_0^2\left(\overline{t^2}-\bar{t}\cdot\bar{t}\right)\right)}}}-z_{0.975}\right)$$

$$where\ \bar{t}\equiv\frac{\sum_{j=0}^{J}t_j}{J+1}\ \&\ \overline{t^2}\equiv\frac{\sum_{j=0}^{J}t_j^2}{J+1}$$

[1]Lu et al. Statistics in Medicine. 2009, 28:679-699

[2]Hu, Mackey, Thomas. In preparation. 2018

	MMRM	RCRM
Model	• Changes over time are unconstrained	• Constant rate of decline w/in each subject
Summary Measure	• Change from baseline → RR • 1° analysis based on last time-point	• Rate of decline → RR • 1° analysis summarizes all data
Time	• Time modeled as visits • Unscheduled visits not used or mapped to scheduled visit	• Time modeled directly • Unscheduled visits easily incorporated
Variability	• Inter- & Intra- variability not separated out • Cannot assess endpoint 'noisiness' *vs* subject heterogeneity	•Inter- & Intra- variability estimated separately •Can assess endpoint 'noisiness' vs subject heterogeneity
Robustness	Type I error • Controlled for MAR Bias • Unbiased under MAR & correct variability assumptions	Type I error • Controlled for MAR under linearity • Controlled under (some types of?) non-linearity[1] Bias • Unbiased for rate & Δ from baseline under MAR with correct assumptions on linearity & variability • Estimates "average" rate of decline under non-linearity
Power	• Sample size • Placebo decline at time T • Study duration • Within/between subject variability	• Sample size • Placebo decline at time T • Study duration • Within/between subject variability • Number and placement of assessments • Common close data adds additional power

[1]Chen et al. CTAD 2014

RCRM

subject i's baseline

subject *i*'s rate of decline (constant over time)

baseline

placebo rate

rate Δ due to tx

assessment time for patient *i* at visit *j*

$$Y_{ij} = (\alpha + u_{0i}) + (\beta + \beta_{tx} \times I_{\{tx\}} + u_{1i}) \cdot time_{ij} + \varepsilon_{ij}$$

$i = 1, \ldots, m + n$
$j = 0, 1, \ldots, J$
$T \equiv time_{ij}$

subject i's outcome at visit *j*

inter-subject baseline variation (σ_0)

inter-subject rate variation (σ_1)

intra-subject variation (σ)